PROCEDURAL ROAD NETWORK GENERATION · REVIEW FOR AV TRAINING DATA

# Roadgen AV Readiness Review

The pipeline builds valid OpenDRIVE networks, but they are not yet accurate or rich enough to train self-driving models on. A quarter of all junctions are really road bends, and those bends cannot be driven at the posted speed. The maps also lack signals, signs, crosswalks, sidewalks and shoulders, the traffic does not react to anything, and there is no export in a format a model can train on.

Based on the code on branch `pre-generation-maps`, measurements of the example networks in `out/examples/`, and the nine paper summaries in `files/`. October 2026.

## What the measurements show

**24–28 %**
OF JUNCTIONS

### Bends are built as junctions

Every node with two roads becomes an OpenDRIVE junction: 119 of 489 in the coastal city, 167 of 605 in the generated-maps city. A lane-graph model trained on these maps learns intersections where there are only curves.

**12 m**
MEDIAN BEND RADIUS

### Bends are too tight for their speed limit

The connecting road of a bend keeps the full speed limit of the roads it joins. At that speed a median bend needs 11.6 m/s² of lateral acceleration, more than 1 g, and 77–83 % of bends exceed a comfortable 3 m/s². These figures use each bend's average curvature, so the peaks are higher still.

**1 in 20**
MAP + SEED PAIRS

### Some lsystem networks cannot be built

lsystem grows segments without knowing how much road each junction will trim off. In 1 of 20 tested combinations an edge was shorter than the trims at its two ends, and the OpenDRIVE builder rejected the network.

**G1**

### Curvature jumps at every junction entry

CONTINUITY ONLY

Straight `<line>` roads meet `paramPoly3` connecting roads with matching heading but a sudden change in curvature. Real roads use clothoids, which change curvature gradually, so steering learned from these maps jumps where real steering would not.

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SIGNALS AND OBJECTS

### The HD maps carry no road furniture or rules

Every lane is a driving lane. There are no signals, signs, stop lines, crosswalks, sidewalks, shoulders or curbs, and no right of way at junctions.

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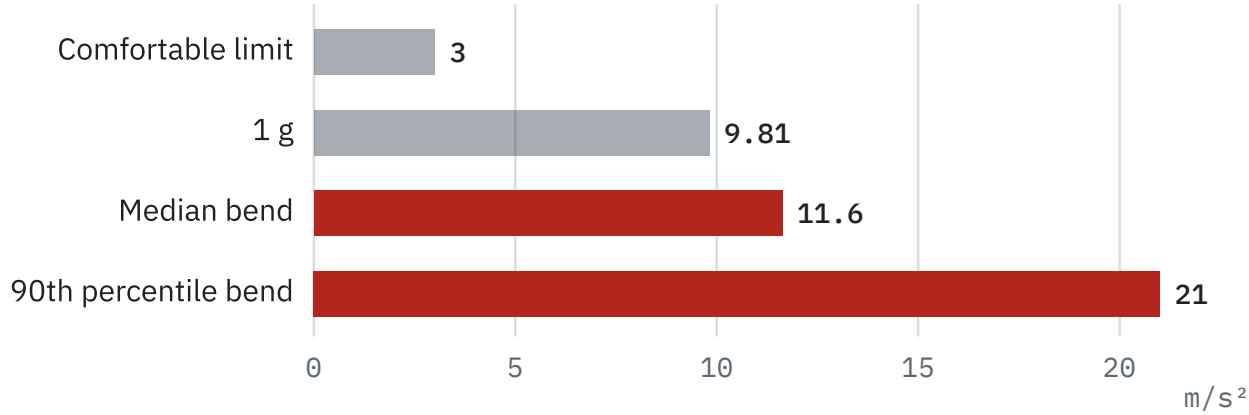
**3**

NEAR-OVERLAPS IN 30 S

### Traffic does not interact

Every vehicle keeps a constant speed under esmini's default controller: no car-following, no yielding and no lane changes. There is also no ego vehicle. This is fine for scalability benchmarks but not as behaviour data.

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Lateral acceleration needed to drive a bend at its posted speed limit, coastal city (374 bend connecting roads). The 90th percentile is 21.3 m/s² there and 20.7 m/s² in the generated-maps city.

## Improvements by priority

A · FIX FIRST

### Geometry accuracy

> **Merge bends into curved roads.**
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> Build each chain of two-road nodes as one road made of line, spiral and arc geometries, with a minimum radius from design speed, $R \geq v^2 / (g \cdot (e + f))$: about 90 m at 50 km/h and 250 m at 80 km/h. Where a bend cannot be widened, store a lower advisory speed. Turn lane-count

changes into lane sections with width tapers instead of junctions.

`StreetGen §2.5` `Muktadir et al. Stage 4`

**Make lsystem aware of junction trims.**

Reject or snap segments that are too short for the trims they will need, with explicit minimum street angle and vertex degree rules. This touches the rule that generators never use OpenDRIVE code, so decide that first; one option is a neutral geometry module both sides share.

`Teng & Bidarra §3.3` `Parish & Müller §3.3.1`

**Grow smooth curved roads, not polylines,**

from tensor-field streamlines, curve smoothing, or a tortuosity target per road.

`Chen et al. §6.1` `Parish & Müller §3.3` `Aliaga et al. §3.1.1`

**Add banking and sloped junctions**

, already listed as next steps in

`docs/maps.md`

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## B HD-map completeness

**Lane types:**

shoulders, sidewalks, curbs, parking, bike lanes and medians next to the driving lanes.

**Road marks:**

a solid yellow centre line, solid lines on junction approaches as a no-lane-change zone, stop lines, and no marks inside junctions.

`Muktadir et al. Stage 4`

**Signals, signs and objects,**

each placed by distance along the road and offset from it: traffic lights with phase plans, stop and yield signs from the road hierarchy with right of way, speed limit signs, and crosswalks at junction borders.

`StreetGen §2.9`

**Junction realism:**

turn lanes and turn restrictions from a lane-link solver, and corner radii by road class (about 3, 4.9 and 7.6 m in Paris).

`Muktadir et al. Stage 3` `StreetGen §2.3–2.5`

## C Long-tail diversity

**Junction variety:**

3 to 7 legs with skew and offset, giving Y, K and five-way shapes. Muktadir et al. motivate their whole paper with a crash at a lane split.

`Muktadir et al. Stage 1, 3`

**Missing road types:**

roundabouts; highway on- and off-ramps, merges, lane drops and lane adds; overpasses, which the planar graph cannot represent today; tunnels, U-turns and cul-de-sacs.

`Yang et al. §4` `Teng & Bidarra §3.1`

**Style mixing inside one map:**

blending grid, radial and organic patterns with control maps; Paris-style radial roads and San Francisco-style contour roads; rotation fields for skewed intersections; a single grid-to-organic knob; a connectivity knob from grid to cul-de-sac suburbs.

`Parish & Müller §3.2.2` `Chen et al. §5.3` `TownSim §3.3` `Lechner et al. §3`

## D Measure diversity and realism

**Network statistics:**

intersections and dead ends per km², connected-node ratio, street density, orientation entropy, curvature distribution and block sizes.

`Yang et al. §7` `TownSim §4.2` `Teng & Bidarra §4.2`

**Per-junction difficulty:**

field of view, maximum curvature, corner deviation, conflict area and an overall complexity score.

`Muktadir et al., expressive range analysis`

**Compare against real cities**

using only their statistics, never their geometry, which keeps the project's no-OpenStreetMap rule. The same numbers let you generate until every junction-type and curvature bin is filled, and order training data by difficulty.

`Aliaga et al. §5`

## E Turn the output into training data

Decide which model the data feeds, because it changes what matters. For **planning and prediction**, esmini is enough: export the vector map and agent tracks in a standard format such as Argoverse 2, Waymo Motion or ScenarioNet, so existing training code works unchanged. For **perception**, esmini's rendering is too simple: use the `.xodr` and the building meshes as assets in CARLA, which imports OpenDRIVE directly, or RoadRunner, and randomise weather, lighting and materials there.

**Ground truth:**

esmini writes OSI ground truth (

`--osi_file`

) with lane boundaries and objects per frame, usable as labels.

**Ego scenarios:**

an ego vehicle with routes chosen for specific manoeuvres (unprotected left turns, merges, roundabout entries), reactive traffic through esmini's ACC controller or SUMO co-simulation, pedestrians at crosswalks, and parametrised OpenSCENARIO with selection of critical cases by time-to-collision.

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## Engineering

**Run history reproducibility:**

record the git commit, whether the working tree had uncommitted changes, and package versions with each run, and optionally hashes of input file contents, which are referenced only by path today.

**Validation:**

the XSD check is always skipped without the ASAM schema. The open-source ASAM Quality Checker for OpenDRIVE could replace it and adds semantic checks.

**Housekeeping:**

`project-explanation.md`

still describes OpenStreetMap networks; 165 ruff warnings mostly conflict with the required

`Optional[...]`

style; the known failing benchmark test needs one of the two fixes in

`docs/development.md`

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## Where to start

1. **Curved roads with design-speed radii.** It fixes the largest measured error and makes everything downstream more realistic.
2. **Lane types, road marks, signals and signs,** the minimum for an HD map that perception and planning models expect.
3. **A metrics command,** so every later change can be measured against real-city statistics.
4. **A training-data export** for the model you are targeting, with OSI ground truth.
5. **Junction and road-type variety,** guided by those metrics.

## Papers drawn on

| PAPER | YEAR | MAIN IDEA USED HERE |
|---|---|---|
| Parish & Müller, Procedural Modeling of Cities | 2001 | Pattern rules and blending, local constraints, curve smoothing |
| Lechner et al., Procedural City Modeling | c. 2003 | Connectivity knob (detour ratio), grid deviation |
| Chen et al., Interactive Procedural Street Modeling | 2008 | Tensor-field curved roads, rotation fields for skewed junctions |
| Aliaga et al., Interactive Example-Based Urban Layout Synthesis | 2008 | Learning spacing, angle and tortuosity statistics from real cities |
| Yang et al., Urban Pattern | 2013 | Layout metrics with real-map reference values, cul-de-sac templates |
| Cura et al., StreetGen | c. 2016 | Corner radii, width tapers, street objects, crosswalks |
| Teng & Bidarra, Semantic patch-based generation | 2017 | Tagged patch library, connect and clip rules, connectivity metrics |
| Song & Whitehead, TownSim | 2019 | Grid-to-organic snap knob, orientation and curvature metrics |